

# Department of Computer Science (https://

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## Jay Bagga

Professor of Computer Science

**Curriculum Vitae** ([https://www.bsu.edu/-/media/www/departments/content/computerscience/cv/cs-vita-2023/jb\\_cv-2024.pdf?sc\\_lang=en](https://www.bsu.edu/-/media/www/departments/content/computerscience/cv/cs-vita-2023/jb_cv-2024.pdf?sc_lang=en))

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Dr. Jay Bagga is a Professor of Computer Science, with over 40 years of experience in research and teaching in a wide variety of graduate and undergraduate topics in Computer Security, Software Verification, Graph Theory and Graph Algorithms, Theory of Computers, Algorithm Design and Complexity, Bioinformatics, Computational Geometry, Programming, and Discrete Structures and Data Structures. He holds Ph. D. and master's degrees in mathematics from Purdue University, and a master's degree in computer science from Ball State University.

Dr. Bagga has published two books, a book chapter, and many research papers. He has presented lectures within U.S. and in many countries abroad. He has collaborated in research with scholars from many universities in US and abroad, and guided Ph. D. students. Dr. Bagga has been awarded several Federal, State, University, and Industry research grants. He has organized national and international professional conferences, and has directed several exchange programs between Ball State University and international institutions. He serves as Editor or Referee for many national and international professional journals and conference proceedings.

Dr. Bagga has received several honors and awards including Ball State Summer Commencement Speaker (2012), Outstanding Faculty Award (Ball State University (2010), Third Frank Harary Memorial Lecture Speaker (New Mexico State University 2009), Researcher of the Year (Ball State University, 2002), and Fulbright Senior Scholar and Visiting Professor (University of Zimbabwe, 2000). Outside of his professional activities, Dr. Bagga enjoys traveling, reading, and spending time with grandchildren.

## Education

- M.S., Computer Science, Ball State University, 1988
- Ph.D., Mathematics, Purdue University, 1984
- M.S., Mathematics, Purdue University, 1977

## Awards

- Mira TechPoint Technology in Education Excellence & Innovation Finalist, 2012
- Ball State Summer Commencement Speaker, 2012
- Outstanding Faculty Award, Ball State University, 2010
- Third Harary Memorial Lecture Speaker, Department of Computer Science, New Mexico State University, 2009
- Harary Plenary Lecture Speaker, MIGHTY (Midwest Graph Theory Conference), Indiana University Purdue University Fort Wayne, 2006
- Researcher of the Year, Ball State University, 2002
- Fulbright Senior Scholar and Visiting Professor, 2000 (University of Zimbabwe)
- Foundation Fellow of the Institute of Combinatorics & its Applications, 1992

## VSTOP Honors and Awards

- 2021 US Election Assistance Commission Clearinghouse Award winner: Outstanding Innovations in Elections – Large Jurisdictions.

- 2021 US EAC Clearie Honorable Mention: Outstanding Use of Help America Vote Act (HAVA) Grants in Election Modernization.

## Additional Responsibilities

- Director, South Asia Exchange Program
- Co-Director, State of Indiana Voting Systems Technical Oversight Program (VSTOP)  
(<http://bowencenterforpublicaffairs.org/institutes/policy-research/election-admin/vstop>)
- Faculty Advisor, Indian Students Association

## Sampling of Recent Scholarly Work

### BOOKS and book chapters

#### BOOK CHAPTER

Disruptive Innovations in Voting: Electronic Poll Book Adoption and Certification in Indiana coauthored with Chad Kinsella and Bryan Byers, in the book **Government Response to Disruptive Innovation: Perspectives and Examinations** (<https://www.igi-global.com/book/government-response-disruptive-innovation/301601>), Editors: Sam B. Edwards, III and James R. Masterson , Projected publication date: March, 2023, IGI Global Publishers, ISBN13: 9781668464298.

#### BOOK

**Line Graphs and Line Digraphs** (<https://doi.org/10.1007/978-3-030-81386-4>), coauthored with Lowell W. Beineke, Springer, 2021

#### BOOK

**Theoretical Computer Science and Discrete Mathematics, First International Conference, ICTCSDM 2016, 2017** (<https://doi.org/10.1007/978-3-319-64419-6>), co-edited with S. Arumugam, Lowell W. Beineke, and B.S. Panda.

## PUBLICATIONS - REFEREED

### JOURNAL ARTICLE

**Big data from a popular app reveals that shing creates superhighways for aquatic invaders** (<https://doi.org/10.1093/pnasnexus/pgac075>), with Jessica L Weir, Kirsten Vacuraa, Adam Berland, Kieran Hyder, Christian Skov, Johan Attbyg, and Paul A Venturelli, PNAS Nexus, 2022, 1, 1{9, <https://doi.org/10.1093/pnasnexus/pgac075>Big data from a popular app reveals that shing creates superhighways for aquatic invaders, with Jessica L Weir, Kirsten Vacuraa, Adam Berland, Kieran Hyder, Christian Skov, Johan Attbyg, and Paul A Venturelli, PNAS Nexus, 2022, 1, 1{9

### JOURNAL ARTICLE

**New results and open problems in line graphs** (<https://doi.org/10.1080/09728600.2022.2093146>), with Lowell W. Beineke, AKCE International Journal of Graphs and Combinatorics, 022

### JOURNAL ARTICLE

**A survey of line digraphs and generalizations** (<https://doi.org/10.47443/dml.2021.s109>), Jay S. Bagga and Lowell W. Beineke, Discrete Math. Lett. 6 (2021) 68{83

### JOURNAL ARTICLE

Graphoidal graphs and graphoidal digraphs: a generalization of line graphs, with S. Arumugam, AKCE International Journal of Graphs and Combinatorics, 2020.

## JOURNAL ARTICLE

A survey and a new class of graceful unicyclic graphs, with Max Pambe Biatch', S. Arumugam, AKCE International Journal of Graphs and Combinatorics, 2020.

## JOURNAL ARTICLE

Discovery of Some New Classes of Graceful Unicyclic Graphs, with L. Fotso and Max Pambe, Journal of Combinatorial Mathematics and Combinatorial Computing, Volume 109, 2019, 105-127.

## JOURNAL ARTICLE

Bagga, J., Beineke, L. W., Varma, B. (2016). A number theoretic problem on super line graphs. AKCE International Journal of Graphs and Combinatorics/Elsevier, 2016(13), 14.

## JOURNAL ARTICLE

Bagga, J. S., Fotso, L. P., Max, P. B., Arumugam, S. (2015). New Classes of Graceful Unicyclic Graphs. Electronic Notes in Discrete Mathematics/Elsevier, 48, 27-32.

## JOURNAL ARTICLE

Begum, M., Bagga, K. S., Blakey, C. A., Saha, S. (2014). Network motif identification and structure detection with graphical models. Network Biology, 14(4), 155-169.

## JOURNAL ARTICLE

Bagga, K. S., Max, P., Fotso, L. (2014). An Algorithm for graceful labelings of certain unicyclic graphs. VNU Journal of Science: Comp. Science & Com. Eng. Vietnam National University, 30(3), 1-11.

## EXTERNAL GRANTS

"Voting Systems Technical Oversight Program", Indiana Secretary of State, 2007- 2023

"Theory of Computation," Sponsored by Vietnam Educational Foundation, 2013 -2014.

"Pre-election Logic and Accuracy Testing & Post Election Audit Initiative," Sponsored by U. S. Election Assistance Commission, 2012 - 2013.

"Design and Analysis of Algorithms - Mobile Applications," Sponsored by Vietnam Education Foundation, 2012 - 2013.

"U. S. Faculty Scholar Program Grant", Sponsored by Vietnam Education Foundation, 2011- 2012

"Curriculum Development for Careers in Homeland Security", Indiana Department of Homeland Security, 2008

"Science Literacy Project," Department of Energy, 2006-2008

## Related Links

**Faculty-maintained web site** (<http://www.cs.bsu.edu/homepages/jay/>)

**LinkedIn** (<https://www.linkedin.com/in/jay-bagga-0721118/>)

## COURSE SCHEDULE

| Course                                                                                                                                                                                                                                                                                          | No. | Section | Times       | Days | Location        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------|-------------|------|-----------------|
| <b>Network Security</b> ( <a href="https://banner.bsu.edu/ssbprod/bwckctlg.p_disp_course_detail?cat_term_in=202420&amp;subj_code_in=cs&amp;crse_numb_in=447">https://banner.bsu.edu/ssbprod/bwckctlg.p_disp_course_detail?cat_term_in=202420&amp;subj_code_in=cs&amp;crse_numb_in=447</a> )     | 447 | 1       | 1100 - 1215 | T R  | RB, room 104    |
| <b>Applications of Comp</b> ( <a href="https://banner.bsu.edu/ssbprod/bwckctlg.p_disp_course_detail?cat_term_in=202420&amp;subj_code_in=cs&amp;crse_numb_in=681">https://banner.bsu.edu/ssbprod/bwckctlg.p_disp_course_detail?cat_term_in=202420&amp;subj_code_in=cs&amp;crse_numb_in=681</a> ) | 681 | C01     | 0000 - 0000 |      | OL, room ONLINE |

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